A New CRT-based Fully Homomorphic Encryption

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Abstract. We propose a novel Fully Homomorphic Encryption (FHE) scheme based on the Chinese Remainder Theorem (CRT), designed to mitigate noise growth during homomorphic operations. By uniquely embedding plaintext and noise into CRT-structured ciphertexts, our scheme supports multiple homomorphic operations without requiring bootstrapping. We provide a rigorous security analysis, proving IND-CPA security under the standard Ring Learning with Errors (RLWE) assumption, thus ensuring robustness against both classical and quantum attacks. We also establish accuracy through formal proofs and recommend secure parameters that balance performance and security, facilitating practical implementation.

Keywords: Fully Homomorphic Encryption (FHE), Chinese Reminder Theorem, Chosen Plaintext Attack

1 Introduction

Fully Homomorphic Encryption (FHE) is a transformative cryptographic paradigm that enables computations on encrypted data without requiring decryption. Introduced by in 1970's by Rivest [18] and first secure scheme was proposed Craig Gentry in 2009 [13] [14], and later improved in [3] [2] [4] [8] [19] [12], FHE has opened new avenues for secure data processing, allowing operations to be performed on ciphertexts while maintaining the confidentiality of the underlying plaintexts. This capability is particularly valuable in scenarios where data privacy is paramount, such as in cloud computing, medical data processing, and financial transactions.

FHE schemes can execute a wide range of operations on encrypted data, including both arithmetic (addition and multiplication) and logical operations, making them Turing complete. This means that any computable function can, in theory, be evaluated on encrypted data without revealing the data itself. The potential applications of FHE are vast, encompassing secure voting systems, privacy-preserving data analysis, and encrypted search functionalities, among others.

One of the critical challenges addressed by FHE is the need to maintain data privacy while enabling the functionality of modern data analytics and machine learning. Traditional encryption schemes secure data at rest and in transit but require decryption for processing, exposing sensitive information to potential breaches. FHE, on the other hand, keeps data encrypted throughout the processing life-cycle, thus significantly enhancing security and privacy.

Despite its promising features, the practical deployment of FHE has been historically hindered by performance issues, particularly the high computational overhead associated with homomorphic operations. Early FHE schemes required bootstrapping [11] [7] [15]—a process to refresh ciphertexts to manage noise growth during computations—which was computationally expensive and impractical for real-world applications [1].

The concept of utilizing the Chinese Remainder Theorem (CRT) for constructing a fully homomorphic encryption (FHE) scheme was first introduced as a "privacy homomorphism" in 1978 by Rivest, Adleman, and Dertouzos [18]. The fundamental idea was to define an encryption function that allows computations on encrypted data without needing to decrypt it. The approach begins with enabling basic binary operations, such as addition and multiplication, over encrypted data. Since any function can be approximated by a polynomial, supporting addition and multiplication on encrypted data implies the potential to compute any function on the data.

The privacy homomorphism designed by Rivest, Adleman, and Dertouzos operates as follows:

- *Key Generation*: The user selects two large prime numbers, p and q, and computes the public parameter $n = pq$.
- *Plaintext and Ciphertext Spaces*: The plaintext space is defined as $\mathbb{Z}_n$, and the ciphertext space is defined as $\mathbb{Z}_p \times \mathbb{Z}_q$.

Encryption:

$$m \in \mathbb{Z}_n \longrightarrow (m \mod p,\ m \mod q) = (c_1, c_2)$$

Decryption:

The plaintext m is recovered from (c_1, c_2) using the Chinese Remainder Theorem.

However, it was later demonstrated that this privacy homomorphism is vulnerable to known plaintext attacks [6]. The attack exploits the fact that if $\mathrm{Enc}(m) = (c_1, c_2)$, then:

$$c_1 = m \mod p \Rightarrow m = pk + c_1 \Rightarrow p|(m - c_1)$$

Since $p|n$, we have:

$$p|\gcd(m - c_1, n)$$

Similarly, $q|\gcd(m - c_2, n)$. Given that $n = pq$, the greatest common divisor $\gcd(m - c_i, n)$ for $i = 1, 2$ can be either p or q. Consequently, an attacker can recover the secret keys from a known set of plaintext and ciphertext values.

This vulnerability highlighted the need for more secure methods of constructing fully homomorphic encryption schemes, leading to the exploration and development of more robust approaches in the following decades.

Our Contribution: In this paper, we propose a novel fully homomorphic encryption (FHE) scheme designed to enable efficient and reliable computations on encrypted data. The core idea of our scheme involves uniquely encoding plaintext with noise in a way that effectively prevents the noise from accumulating excessively and corrupting the underlying data. This innovative encoding approach ensures that the integrity and accuracy of computations on encrypted data remain intact, addressing the common issue of noise growth that typically hampers FHE schemes.

This paper aims to provide a comprehensive overview of the proposed Fully Homomorphic Encryption, discussing its foundational principles, security analysis, speed, performance, and practical applications. We will also explore the implementation and benchmarking of the proposed scheme.

We reduce the security of the proposed scheme to lattice-based cryptography, meaning it is as secure as any other lattice-based cryptographic scheme. Specifically, we demonstrate that if a method exists to break the proposed scheme, the same method can be used to break the known hard problem, LWE and RLWE problem. Therefore, our proposed FHE scheme is as secure as any cryptographic scheme based on the LWE problem, which is recognized as being quantum-safe.

The rest of the section is organized as follows: we begin with an introduction to the preliminaries, followed by the definition of essential notations and parameters. Next, we describe the proposed fully homomorphic encryption (FHE) algorithm in detail. Following this, we provide a proof of the algorithm's correctness and conduct a thorough security analysis. Additionally, we present a performance analysis and benchmarking results.

2 Preliminary

In this section, we provide an overview of the basic concepts that underpin our Fully Homomorphic Encryption (FHE) scheme, focusing on the Chinese Remainder Theorem (CRT). These concepts are fundamental to understanding the security and functionality of our encryption scheme.

2.1 Notation

Let n be a power of two and $f(X) = X^n + 1$. Set $\mathcal{R} := \mathbb{Z}[X]/\langle f \rangle$ and, for $q \geq 2$, $\mathcal{R}_q := \mathbb{Z}_q[X]/\langle f \rangle$. We also use $\mathbb{Z}$ when scalars are clear from context. For $a \in \mathbb{Z}$ or $a \in \mathcal{R}$, write $[a]_q$ for the centered reduction modulo q, i.e., coefficients in $(-q/2, q/2]$. For coprime p_1, p_2, define $\varDelta_1 := p_2\,[p_2^{-1}]_{p_1}$ and $\varDelta_2 := p_1\,[p_1^{-1}]_{p_2}$, and $CRT_{p_1,p_2}(x, y) := (\varDelta_1 x + \varDelta_2 y) \bmod p_1 p_2$, which satisfies $CRT_{p_1,p_2}(x, y) \bmod p_1 = x$ and $\bmod p_2 = y$. When $x, y \in \mathcal{R}$ (or vectors), CRT_{p_1,p_2} acts coefficientwise. The Euclidean norm $\|\cdot\|$ of a vector $\mathbf{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$ is denoted as $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$.

Parameters. λ is the security parameter; $n = n(\lambda)$; $q = q(\lambda)$ is the ciphertext modulus; $\chi = \chi(\lambda)$ is the RLWE noise distribution. Auxiliary moduli p_1, p_2 are pairwise coprime with q and satisfy $p_1 p_2 < q/2$. $PP = (\lambda, n, q, \chi, p_1, p_2, \mathcal{R}_q)$

2.2 Theoretical Backgorund

The Chinese Remainder Theorem (CRT) is a key mathematical tool used in number theory and cryptography. It provides a way to solve systems of simultaneous congruences with pairwise co-prime modulo.

Theorem 1 (Chinese Remainder Theorem). *Let $n_1, n_2, \ldots, n_k$ be pairwise coprime integers (i.e., $\gcd(n_i, n_j) = 1$ for $i \neq j$). For any given integers $a_1, a_2, \ldots, a_k$, there exists an integer x that simultaneously satisfies the system of congruences:*

$$x \equiv a_i \mod n_i, \quad i = 1, 2, ..., k$$

Moreover, the solution x is unique modulo $N = n_1 n_2 \cdots n_k$.

The CRT is useful in cryptographic applications because it allows computations to be performed independently in smaller, modular arithmetic spaces and then recombined to form the final result. This can lead to efficiencies in both computation and storage.

Definition 1 (CRT Function). *For pairwise coprime integers $p_1, \ldots, p_k$, we define the CRT function $CRT_{(p_1,\ldots,p_k)}$ for inputs $(m_1, \ldots, m_k)$ as a number in $\mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k}$ which is congruent to m_i modulo p_i for all $i \in \{1, \ldots, k\}$, where $p = \prod_{i=1}^{k} p_i$.*

Formally, we have:

$$CRT_{(p_1,\ldots,p_k)}(m_1, \ldots, m_k) \equiv \sum_{i=1}^{k} m_i \hat{p}_i (\hat{p}_i^{-1} \mod p_i) \mod p,$$

where

$$\hat{p}_i = \frac{p}{p_i} = \prod_{\substack{j=1 \\ j \neq i}}^{k} p_j$$

Further for $\mathbf{a}, \mathbf{b} \in \mathbb{Z}[X]/\langle X^n + 1 \rangle$ wherever $\mathbf{a} = a_0 + a_1 x + ... + a_{n-1} x^{n-1}$, $\mathbf{b} = b_0 + b_1 x + ... + b_{n-1} x^{n-1}$ and consequently we denote

$$\begin{aligned} CRT_{p_1,p_2}(\mathbf{a}, \mathbf{b}) = {} & CRT_{p_1,p_2}(a_0, b_0) + CRT_{p_1,p_2}(a_1, b_1) x + \ldots \\ & + CRT_{p_1,p_2}(a_{n-1}, b_{n-1}) x^{n-1} \end{aligned}$$

The ring learning with errors (RLWE) problem was introduced by Lyubashevsky, Peikert, and Regev in [16]. We will use a simplified special-case version of the problem that is easier to work with [3].

Definition 2 (Decision Ring-LWE Problem). *Let $f(X) = X^n + 1$ (with n a power of two) and define the cyclotomic ring $\mathcal{R}_q = \mathbb{Z}_q[X]/\langle f(X)\rangle$. Fix a noise distribution χ over $\mathcal{R}_q$ and draw a secret $s \xleftarrow{\$} \mathcal{R}_q$. The* Ring-LWE distribution *$\mathcal{A}_{s,\chi}$ on $\mathcal{R}_q \times \mathcal{R}_q$ is*

$$\mathcal{A}_{s,\chi} := \left\{ \big(a,\ a \cdot s + e\big) \;\middle|\; a \xleftarrow{\$} \mathcal{R}_q,\ e \xleftarrow{\$} \chi \right\}.$$

For any probabilistic polynomial-time (PPT) distinguisher $\mathcal{D}$ we define its advantage *in the average-case decision Ring-LWE problem as*

$$Adv_{\mathcal{D}}^{\mathrm{RLWE}_{n,q,\chi}} := \left| \Pr_{s,a,e}\big[\mathcal{D}\big(a, a \cdot s + e\big) = 1\big] - \Pr_{a,b \leftarrow \mathcal{R}_q^2}\big[\mathcal{D}(a,b) = 1\big] \right|.$$

The decision Ring-LWE problem $\mathrm{RLWE}_{n,q,\chi}$ *is said to be* hard *if $Adv_{\mathcal{D}}^{\mathrm{RLWE}_{n,q,\chi}}$ is negligible in the security parameter for every PPT distinguisher $\mathcal{D}$.*

The RLWE problem is important because it enables reductions from well-known hard lattice problems, such as the shortest vector problem (SVP) over ideal lattices.

Theorem 2 (Lyubashevsky-Peikert-Regev [16]). *For any d that is a power of 2, define the ring $\mathcal{R} = \mathbb{Z}[x]/\langle x^d + 1\rangle$. Let $q = q(d) \equiv 1 \mod d$ be a prime integer, and let $B = \omega(\sqrt{d}\log d)$. There exists an efficiently samplable distribution χ that outputs elements of $\mathcal{R}$ of length at most B with overwhelming probability. If there exists an efficient algorithm that solves $RLWE_{d,q,\chi}$, then there is an efficient quantum algorithm for solving $d^{\omega(1)} \cdot (q/B)$-approximate worst-case SVP for ideal lattices over $\mathcal{R}$.*

Typically, to use RLWE in cryptographic applications, the noise distribution χ is chosen to be Gaussian. Vectors sampled from this distribution have norm bounded by $\mathrm{poly}(d)$ with overwhelming probability. In certain cases, the Gaussian distribution may need to be "ellipsoidal" for reductions to apply [16]. Prior work has shown that for RLWE, one can equivalently assume that the secret s is sampled from the noise distribution χ [16].

3 Construction

In this section, we outline the construction of our Fully Homomorphic Encryption (FHE) scheme, focusing on its key components: key generation, encryption, homomorphic operations, and decryption. Our construction builds on the principles of lattice-based cryptography, incorporating enhancements to achieve improved performance and security. First we describe the required parameters for the scheme along with their sizes, then the actual construction of the scheme and finally we prove the correctness of the scheme. The scheme has 4 major components key generation, encryption, decryption, and Homomorphic operations e.g. adition and multiplication.

The scheme consists of the following algorithms:

CRT-FHE.Setup(1^λ): Takes as input the security parameter λ and sets up global parameters for the scheme.

- Let λ be the security parameter. Define the following public parameters:
  - $q = q(\lambda)$: Ciphertext modulus.
  - $f(X) = X^n + 1$: Cyclotomic polynomial of degree $n = n(\lambda)$.
  - $\mathcal{R}_q = \mathbb{Z}_q[X]/\langle f(X) \rangle$: Polynomial ring modulo $f(X)$ with coefficients in $\mathbb{Z}_q$.
  - $\chi = \chi(\lambda)$: Discrete noise distribution, e.g., a Gaussian or binomial distribution.
- Define auxiliary CRT moduli p_1 and p_2 satisfying:
  - $p_1 \cdot p_2 < q/2$: Ensures correctness under bounded noise growth.
  - $\gcd(p_1, p_2) = \gcd(p_1, q) = \gcd(p_2, q) = 1$: All parameters are pairwise coprime.
- Output public parameters:

$$PP = (\lambda, n, q, \chi, p_1, p_2, \mathcal{R}_q)$$

CRT-FHE.KeyGen(PP): Given the public parameters, generates keys for encryption and evaluation.

- **Input:** Public parameters PP
- **Output:** Public key $pk = (\mathbf{pk}_0, \mathbf{pk}_1)$, secret key $sk = \mathbf{s}$
- **Procedure:**
  1. Sample secret key: $\mathbf{s} \leftarrow \chi$ (see LPR Lemma for security of small secret keys)
  2. Sample $\mathbf{a} \in \mathcal{R}_q$ uniformly at random
  3. Sample noise $\mathbf{e} \leftarrow \chi$
  4. Define CRT recombination:
  $$CRT_{p_1,p_2}(0, \mathbf{e}) = \Delta_2 \cdot \mathbf{e} \mod q, \quad \text{where } \Delta_2 = p_1[p_1^{-1}]_{p_2}$$
  5. Compute:
  $$\mathbf{b} = \mathbf{a} \cdot \mathbf{s} + CRT_{p_1,p_2}(0, \mathbf{e}) \mod q$$
  6. Set:
  $$pk = (\mathbf{pk}_0, \mathbf{pk}_1) = (\mathbf{b}, -\mathbf{a}), \quad sk = \mathbf{s}$$

CRT-FHE.RelinKeyGen(sk): Generates the relinearization key used to reduce ciphertext degree after multiplication.

- Let $w = 2^r$ be the decomposition base, and let $\ell = \lfloor \log_w(q) \rfloor$ denote the number of base-w digits needed to represent ciphertext coefficients.
- For each $i = 0, 1, \ldots, \ell$, perform:
  1. Sample noise polynomials: $\mathbf{e}_i, \mathbf{e}'_i \leftarrow \chi$
  2. Sample uniform polynomial: $\mathbf{a}_i \leftarrow \mathcal{R}_q$

3. Compute:

$$r_{i,0} = [\mathbf{a}_i \cdot \mathbf{s} + w^i \cdot \mathbf{s}^2 + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}_i)]_q$$
$$r_{i,1} = [-\mathbf{a}_i + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'_i)]_q$$

– The evaluation key is:

$$\mathsf{evk} = \{(r_{i,0}, r_{i,1})\}_{i=0}^{\ell}$$

This key is published with the public parameters and is used during relinearization to transform a degree-2 ciphertext back into standard form.

CRT-FHE.Enc(pk, $\mathbf{m} \in \mathbb{Z}_{p_1}^n$): Encrypts a plaintext vector into a degree-1 ciphertext.

– **Input:** Public key $pk = (\mathbf{pk}_0, \mathbf{pk}_1)$, message vector $\mathbf{m} = (m_0, \ldots, m_{n-1}) \in \mathbb{Z}_{p_1}^n$
– **Output:** Ciphertext $\mathbf{c} = (c_0, c_1) \in \mathcal{R}_q^2$
– **Procedure:**
1. Encode plaintext as polynomial in R_{p_1}:

$$\mathbf{m}(x) = m_0 + m_1 x + \cdots + m_{n-1} x^{n-1}$$

2. Sample $\mathbf{u}, \mathbf{e}', \mathbf{e}'' \leftarrow \chi$
3. Compute:

$$c_0 = [\mathbf{pk}_0 \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}')]_q$$
$$c_1 = [\mathbf{pk}_1 \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'')]_q$$

4. Output ciphertext $\mathbf{c} = (c_0, c_1)$

CRT-FHE.Dec(sk, $\mathbf{c} = (c_0, c_1)$): Decrypts a ciphertext to recover the original message.

– **Input:** Secret key $\mathbf{s}$, ciphertext $\mathbf{c} = (c_0, c_1) \in \mathcal{R}_q^2$
– **Output:** Plaintext $\mathbf{m} \in R_{p_1}$
– **Procedure:**
1. Compute:

$$\mathbf{m} = [(c_0 + c_1 \cdot \mathbf{s}) \bmod q] \bmod p_1$$

2. Output $\mathbf{m}$

Structure of Ciphertext The ciphertext can be written as

$$\begin{aligned}
c &= \left([\mathbf{pk}_0 \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}')]_q\,,\ [\mathbf{pk}_1 \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'')]_q\right) \\
&= \left([(\mathbf{a} \cdot \mathbf{s} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e})) \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}')]_q\,,\ [-\mathbf{a} \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'')]_q\right) \\
&= \left([\mathbf{a} \cdot \mathbf{u} \cdot \mathbf{s} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e} \cdot \mathbf{u} + \mathbf{e}')]_q\,,\ [-\mathbf{a} \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'')]_q\right) \\
&= \left([\mathbf{a}' \cdot \mathbf{s} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}''')]_q\,,\ [-\mathbf{a}' + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'')]_q\right) \qquad (1)
\end{aligned}$$

Homomorphic Operations: With the encryption, decryption, and relinearization procedures defined, the CRT-FHE scheme supports essential homomorphic operations on ciphertexts. These include homomorphic addition and multiplication, allowing computations to be performed directly over encrypted data without requiring decryption. Homomorphic addition is a simple component-wise operation on ciphertexts, preserving the degree and enabling addition of plaintexts under encryption. Homomorphic multiplication, on the other hand, increases the ciphertext degree and necessitates a relinearization step to bring the ciphertext back to a standard two-component form suitable for further operations. The following subsections describe these operations in detail.

Homomorphic Addition. Let $\mathbf{c}_1 = (c_0^1, c_1^1)$ and $\mathbf{c}_2 = (c_0^2, c_1^2)$ be two ciphertexts in $\mathcal{R}_q^2$. The homomorphic addition is performed component-wise:

$$\mathbf{c}_{\text{add}} = (c_0^1 + c_1^1 \bmod q,\ c_0^2 + c_1^2 \bmod q)$$

This results in a ciphertext $\mathbf{c}_{\text{add}}$ that encrypts the sum of the plaintexts.

Homomorphic Multiplication Let $\mathbf{c}_1 = (c_0^1, c_1^1)$ and $\mathbf{c}_2 = (c_0^2, c_1^2)$ be ciphertexts encrypting plaintexts $\mathbf{m}_1$ and $\mathbf{m}_2$ respectively. Their product is computed as:

$$\begin{aligned}
\mathbf{c}_{\text{mult}} &= (c_0, c_1, c_2) \in \mathcal{R}_q^3, \text{ where} \\
c_0 &= c_0^1 \cdot c_0^2 \\
c_1 &= c_0^1 \cdot c_1^2 + c_1^1 \cdot c_0^2 \\
c_2 &= c_1^1 \cdot c_1^2
\end{aligned}$$

The resulting ciphertext $\mathbf{c}^*_{\text{mult}}$ encrypts the product $\mathbf{m}_1 \cdot \mathbf{m}_2$, but is of degree 2 in the secret key. To reduce its size and enable further operations, relinearization is applied:

Relinearization Decompose c_2 in base w:

$$c_2 = \sum_{i=0}^{\ell} c_i^2 \cdot w^i$$

Let $\mathsf{evk} = \{(\mathbf{r}_0^i, \mathbf{r}_1^i)\}_{i=0}^{\ell}$ be the evaluation key containing encryption of s^2 scaled by w^i. Then relinearization is computed as:

$$c_0' = c_0 + \sum_{i=0}^{\ell} c_i^2 \cdot \mathbf{r}_0^i$$

$$c_1' = c_1 + \sum_{i=0}^{\ell} c_i^2 \cdot \mathbf{r}_1^i$$

The relinearized ciphertext $\mathbf{c}_{\text{mult}} = (c_0', c_1') \in \mathcal{R}_q^2$ is now of degree 1, encrypting $\mathbf{m}_1 \cdot \mathbf{m}_2$ and compatible with further homomorphic operations.

Noise Growth and Decryption Process.
After relinearization, the ciphertext $\mathbf{c}_{\text{mult}} = (c_0', c_1') \in \mathcal{R}_q^2$ is once again of degree

1, encrypting the product $\mathbf{m}_1 \cdot \mathbf{m}_2$ and compatible with further homomorphic operations. To analyze decryption correctness and track noise growth across operations, we proceed as follows.

The plaintext modulus p is defined as the product of the two CRT moduli:

$$p = p_1 \cdot p_2$$

A general ciphertext component before relinearization takes the following form, representing the linear term plus a multiple of p and the CRT-encoded message with noise:

$$\mathbf{a} \cdot \mathbf{s} + k \cdot p + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}) \mod q$$

During ciphertext multiplication, the CRT encoding of two messages with their respective noise terms multiplies as:

$$\mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}) \cdot \mathrm{CRT}_{p_1,p_2}(\mathbf{m}', \mathbf{e}') = [\mathrm{CRT}_{p_1,p_2}(\mathbf{m} \cdot \mathbf{m}', \mathbf{e} \cdot \mathbf{e}')]_p$$

After multiplication, an additional multiple of p appears in the result to preserve consistency with the CRT domain:

$$\mathrm{CRT}_{p_1,p_2}(\mathbf{m} \cdot \mathbf{m}', \mathbf{e} \cdot \mathbf{e}') + k \cdot p$$

When performing modulus switching or scaling (such as in leveled FHE operations), the message and noise scale accordingly. For example, after scaling by t, the noise grows linearly with t:

$$t \cdot \mathbf{e} + \mathbf{m}$$

Similarly, when rescaling ciphertexts down to a smaller modulus (from q to t), the following approximate scaling relation governs the transformation:

$$\left[\frac{q}{t}\right] \cdot \mathbf{m} + \mathbf{e}$$

After homomorphic multiplication and relinearization, decryption is performed by computing the inner product of the ciphertext with the secret key:

$$c_0 + c_1 \cdot \mathbf{s} = \mathbf{m} + p^r \cdot \mathbf{e} \mod q$$

As the circuit depth r increases, the noise term scales as $p^r \cdot \mathbf{e}$, leading to the following representation for deeper levels:

$$c_0 + c_1 \cdot \mathbf{s} \to p^{e-r} \cdot \mathbf{m} + p^e \cdot \mathbf{e}$$

Finally, after modulus switching and noise accumulation, the ciphertext before final decryption can be expressed as:

$$c_0 + c_1 \cdot \mathbf{s} = \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}) + p^r \cdot \mathbf{e} \mod q$$

This form highlights how the plaintext $\mathbf{m}$, the CRT noise $\mathbf{e}$, and the multiplicative noise term $p^r \cdot \mathbf{e}$ together influence decryption correctness at various levels of homomorphic computation.

3.1 Batch Encoding

Batching is a powerful technique in Fully Homomorphic Encryption (FHE) that allows multiple plaintexts to be encrypted and processed simultaneously within a single ciphertext. This approach significantly enhances the efficiency of homomorphic computations, making FHE more practical for real-world applications. The concept of batching leverages the structure of certain algebraic objects, such as rings and polynomials, to pack multiple messages into one.

Mathematical Foundation To enable coordinate-wise multiplication over plaintext vectors, the underlying polynomial multiplication must be transformed. This is achieved using the Number Theoretic Transform (NTT) and its inverse (INTT), which map the plaintexts into a domain where polynomial multiplication corresponds to element-wise vector multiplication. The following section outlines the mathematical principles supporting this transformation. The current form of the ciphertext:

$$c_0 + c_1 s = \mathbf{m}^* = CRT_{p_1,p_2}(m_0, e_0) + CRT_{p_1,p_2}(m_1, e_1)x + ... + CRT_{p_1,p_2}(m_{n-1}, e_{n-1})x^{n-1}$$

We rewrite it to express the plaintext and noise separately as follows

$$\begin{aligned}
&\mathrm{CRT}_{p_1,p_2}(m_0, e_0) + \mathrm{CRT}_{p_1,p_2}(m_1, e_1) \cdot x + \ldots + \mathrm{CRT}_{p_1,p_2}(m_{n-1}, e_{n-1}) \cdot x^{n-1} \\
&= \sum_{i=0}^{n-1} \mathrm{CRT}_{p_1,p_2}(m_i, 0) \cdot x^i + \sum_{i=0}^{n-1} \mathrm{CRT}_{p_1,p_2}(0, e_i) \cdot x^i \\
&= \sum_{i=0}^{n-1} p_2 \cdot [p_2^{-1}]_{p_1} \cdot m_i \cdot x^i + \sum_{i=0}^{n-1} p_1 \cdot [p_1^{-1}]_{p_2} \cdot e_i \cdot x^i \\
&= \left(p_2 \cdot [p_2^{-1}]_{p_1}\right) \cdot \sum_{i=0}^{n-1} m_i \cdot x^i + \left(p_1 \cdot [p_1^{-1}]_{p_2}\right) \cdot \sum_{i=0}^{n-1} e_i \cdot x^i \\
&= \Delta_1 \cdot \mathbf{m} + \Delta_2 \cdot \mathbf{e}
\end{aligned}$$

Let $\mathbf{c}_1 = (c_0^1, c_1^1)$ and $\mathbf{c}_2 = (c_0^2, c_1^2)$ be two ciphertexts that encrypt plaintexts μ_1 and μ_2, respectively.

$$c_0^j + c_1^j s = \mathbf{m}_i^* = CRT_{p_1,p_2}(m_0^j, e_0^j) + CRT_{p_1,p_2}(m_1^j, e_1^j)x + ... \\ + CRT_{p_1,p_2}(m_{n-1}^j, e_{n-1}^j)x^{n-1}$$

Now when two ciphertexts are multiplied the corresponding polynomials $\sum_{i=0}^{n-1} m_i^1 x^i$ and $\sum_{i=0}^{n-1} m_i^2 x^i$ were multiplied.

$$\text{Dec}(\mathbf{c}_{\text{mult}}, sk) = \mathbf{m}_1^*.\mathbf{m}_2^* = (\sum_{i=0}^{n-1} m_i^1 x^i)(\sum_{i=0}^{n-1} m_i^2 x^i)$$

This does not gives use the coordinate-wise multiplication of the two vectors i.e.

$$(m_0^1, m_1^1, ..., m_{n-1}^1) * (m_0^2, m_1^2, ..., m_{n-1}^2) \rightarrow (m_0^1 \cdot m_0^2, m_1^1 \cdot m_1^2, \ldots, m_{n-1}^1 \cdot m_{n-1}^2)$$

This can be achieved using the concept of NTT as follows:

$$\begin{aligned}
\text{INTT}(m_0^1, m_1^1, \ldots, m_{n-1}^1) &= \boldsymbol{\mu}_1 = \mu_{1,0} + \mu_{1,1}x + \cdots + \mu_{1,n-1}x^{n-1} \\
\text{INTT}(m_0^2, m_1^2, \ldots, m_{n-1}^2) &= \boldsymbol{\mu}_2 = \mu_{2,0} + \mu_{2,1}x + \cdots + \mu_{2,n-1}x^{n-1} \\
\boldsymbol{\mu}_1 \cdot \boldsymbol{\mu}_2 &= \text{INTT}(m_0^1, m_1^1, \ldots, m_{n-1}^1) \cdot \text{INTT}(m_0^2, m_1^2, \ldots, m_{n-1}^2) \\
&= \text{INTT}(m_0^1 \cdot m_0^2, m_1^1 \cdot m_1^2, \ldots, m_{n-1}^1 \cdot m_{n-1}^2)
\end{aligned}$$

In order to achieve this instead of directly putting plaintext values in coefficient, we need to compute the INTT of the plaintext vector and get a polynomial and we encrypt the new polynomial.

Encoding To batch multiple plaintext messages $\mathbf{m} = (m_0, m_1, \ldots, m_{n-1})$, each $m_i \in \mathbb{Z}_{p_1}$, into a single polynomial, we perform the following steps:

– Compute the Inverse NTT of the plaintext as

$$\mu = INTT(\mathbf{m}) = \mu_0 + \mu_1 x + \ldots + \mu_{n-1}x^{n-1}$$

The $INTT()$ function is performed in the space Z_{p_1} so that the result $\mu \in Z_{p_1}[X]$.

– Then the new polynomial is encoded into the following polynomial.

$$\mu^* = CRT_{p_1,p_2}(\mu_0, 0) + CRT_{p_1,p_2}(\mu_1, 0)x + ... + CRT_{p_1,p_2}(\mu_{n-1}, 0)x^{n-1}$$

Decoding To decode the batched ciphertext back into individual plaintexts:

– Apply $\mod p_1$ to separate plaintext and noise.

$$\mu^* \mod p_1$$

$$= CRT_{p_1,p_2}(\mu_0, e_0) + CRT_{p_1,p_2}(\mu_1, e_1)x + ... + CRT_{p_1,p_2}(\mu_{n-1}, e_{n-1})x^{n-1} \mod p_1$$

$$= \mu_0 + \mu_1 x + \ldots + \mu_{n-1}x^{n-1} = \mu$$

– Use the NTT function to transform the polynomial in $\mathbb{Z}_{p_1}[X]$ back into a tuple of elements in $\mathbb{Z}_{p_1}$.

$$NTT(\mu) = \mathbf{m} = (m_0, m_1, ..., m_{n-1})$$

– Extract the individual plaintext messages from the resulting polynomial.

Homomorphic Operations on Batches The primary advantage of batching is that homomorphic operations on the ciphertexts can be performed in parallel on the individual plaintexts. For instance:

Addition:

– Given two batched ciphertexts c_1 and c_2 encoding plaintexts $\mathbf{m_1} = m_1^1, m_2^1, \ldots, m_k^1)$ and $\mathbf{m_2} = (m_1^2, m_2^2, \ldots, m_k^2)$, respectively, their homomorphic addition results in a ciphertext encoding $\mathbf{m_1} + \mathbf{m_2} = (m_1^1 + m_1^2, m_2^1 + m_2^2, \ldots, m_k^1 + m_k^2)$.

Multiplication:

– Similarly, homomorphic multiplication of two batched ciphertexts results in a ciphertext encoding the product of the corresponding plaintexts:

$$\mathbf{m_1} \times \mathbf{m_2} = (m_1^1 \times m_1^2, m_2^1 \times m_2^2, \ldots, m_k^1 \times m_k^2)$$

Batching in FHE schemes offers a powerful means to enhance the efficiency of homomorphic computations by leveraging the parallelism in polynomial rings. By packing multiple plaintexts into a single ciphertext, batching allows for significant improvements in computational and memory efficiency, making FHE more practical for a wider range of applications.

3.2 Correctness

In this section, we prove that the encryption and decryption algorithms correctly recover the original plaintext under bounded noise. We begin with the correctness of decryption for a fresh ciphertext.

Theorem 3. *Let $\mathbf{m} \in R_{p_1}$ be a plaintext and let $(\mathbf{c}_0, \mathbf{c}_1) = Enc(\mathbf{m}, pk)$ be the ciphertext. Then decryption using the secret key $\mathbf{s}$ recovers $\mathbf{m}$, i.e.,*

$$Dec((\mathbf{c}_0, \mathbf{c}_1), \mathbf{s}) = \mathbf{m}$$

provided the accumulated noise satisfies $\|\mathbf{e}^*\|_\infty < \frac{q}{2p_1}$.

Proof. The encryption of $\mathbf{m}$ is computed as:

$$\begin{aligned}
\mathbf{c}_0 &= \mathbf{pk}_0 \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}') \\
&= (\mathbf{a} \cdot \mathbf{s} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e})) \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}') \\
&= \mathbf{a} \cdot \mathbf{u} \cdot \mathbf{s} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e} \cdot \mathbf{u}) + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}') \\
&= \mathbf{a}' \cdot \mathbf{s} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}' + \mathbf{e} \cdot \mathbf{u}) \\
\mathbf{c}_1 &= \mathbf{pk}_1 \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'') = -\mathbf{a} \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'')
\end{aligned}$$

The decryption computes:

$$\begin{aligned}
\mathbf{c}_0 + \mathbf{c}_1 \cdot \mathbf{s} \quad &= \big(\mathbf{a}' \cdot \mathbf{s} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}, \mathbf{e}' + \mathbf{e} \cdot \mathbf{u})\big) + \big(-\mathbf{a} \cdot \mathbf{u} \cdot \mathbf{s} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'')\big) \\
&= \mathrm{CRT}_{p_1,p_2}\left(\mathbf{m},\ \mathbf{e}' + \mathbf{e} \cdot \mathbf{u} + \mathbf{e}'' \cdot \mathbf{s}\right) \\
&= \mathrm{CRT}_{p_1,p_2}\left(\mathbf{m},\ \mathbf{e}^*\right)
\end{aligned}$$

where $\mathbf{e}^* = \mathbf{e}' + \mathbf{e} \cdot \mathbf{u} + \mathbf{e}'' \cdot \mathbf{s}$ denotes the accumulated noise polynomial.

To recover $\mathbf{m}$, we reduce the above expression modulo p_1:

$$\mathrm{Dec}((\mathbf{c}_0, \mathbf{c}_1), \mathbf{s}) = [\mathbf{c}_0 + \mathbf{c}_1 \cdot \mathbf{s}]_q \bmod p_1 = \mathbf{m}$$

provided that the coefficients of $\mathbf{e}^*$ remain strictly less than $q/(2p_1)$ in absolute value, so that reduction modulo q followed by modulo p_1 yields the correct plaintext.

3.3 Noise Growth Behavior in CRT Ciphertexts

We analyze the growth of decryption noise for ciphertexts that use the CRT embedding

$$CRT_{p_1,p_2}(m,e) = \Delta_1 m + \Delta_2 e \bmod p, \qquad p = p_1 p_2,\ \ \Delta_1 = p_2[p_2^{-1}]_{p_1},\ \ \Delta_2 = p_1[p_1^{-1}]_{p_2}.$$

Our analysis follows the methodology of [5,9,10], adapted to account for the two-modulus CRT structure.

Rings, norms, and tails. Let $R = \mathbb{Z}[X]/(X^n+1)$ with n a power of two, and $R_q = R/qR$. Denote by $\sigma_1, \ldots, \sigma_n : R \to \mathbb{C}$ the canonical embeddings, and write

$$\|x\|_{\mathrm{can}} = \Big(\sum_{i=1}^{n} |\sigma_i(x)|^2\Big)^{1/2}, \qquad \|x\|_\infty = \max_j |x_j| \text{ (coefficient } \ell_\infty).$$

There exist fully explicit ring-dependent constants $\kappa_{\infty\leftarrow\mathrm{can}}, \kappa_{\mathrm{can}\leftarrow\infty} > 0$ such that for all $x \in R$ (see, e.g., [17]):

$$\|x\|_\infty \le \kappa_{\infty\leftarrow\mathrm{can}} \cdot \|x\|_{\mathrm{can}} \quad \text{and} \quad \|x\|_{\mathrm{can}} \le \kappa_{\mathrm{can}\leftarrow\infty} \cdot \sqrt{n}\,\|x\|_\infty. \tag{2}$$

Throughout we assume that all error terms are drawn independently from a subgaussian distribution χ over R with parameter σ (e.g., discrete Gaussian), and that s, u are either sampled from χ or from a bounded small-secret distribution. For $e \leftarrow \chi$ and any $\alpha \in R$,

$$\Pr[\,|\langle \alpha, e\rangle| > t\,\sigma\|\alpha\|_2\,] \le 2e^{-\pi t^2} \qquad (\forall t > 0). \tag{3}$$

In particular, for any fixed failure probability $\varepsilon \in (0,1)$, with probability at least $1-\varepsilon$ we have

$$\|e\|_{\mathrm{can}} \le \tau(\varepsilon)\,\sigma\sqrt{n} \quad \text{with} \quad \tau(\varepsilon) = \sqrt{2\log\big(2n/\varepsilon\big)}. \tag{4}$$

Critical quantity and centered CRT-error. Given a ciphertext $c = (c_0, c_1) \in R_q^2$, define its *critical quantity*

$$v(c) = \big[c_0 + c_1 s\big]_q \in R_q,$$

where $[\cdot]_q$ denotes coefficient-wise centered reduction to $(-\frac{q}{2}, \frac{q}{2}]$. If c encrypts $m \in R_{p_1}$ under the CRT embedding, then there exists an (integral) *centered CRT-error* $\eta(c) \in R$ such that, as integers,

$$v(c) = \Delta_1 m + \Delta_2 \eta(c) \pmod{q}. \tag{5}$$

We will measure noise via $\|\eta(c)\|_{\mathrm{can}}$, which isolates the contribution that can cause decryption failure.

Lemma 1 (Correctness threshold). *Let $M_\infty := \max_{m\in R_{p_1}} \|\Delta_1 m\|_\infty$. If*

$$\kappa_{\infty\leftarrow\mathrm{can}} \cdot \|\Delta_2 \eta(c)\|_{\mathrm{can}} \ < \ \frac{q}{2} - M_\infty,$$

then decryption of c returns m.

Proof. By (2), $\|x\|_\infty \le \kappa_{\infty\leftarrow\mathrm{can}}\|x\|_{\mathrm{can}}$ for all $x \in R$. If $\|\Delta_1 m + \Delta_2\eta(c)\|_\infty < q/2$, then the centered lift of $v(c)$ equals the integer polynomial $\Delta_1 m + \Delta_2\eta(c)$, hence $v(c) \bmod p = \Delta_1 m + \Delta_2\eta(c) \bmod p$. Reducing modulo p_1 eliminates the Δ_2 component and yields m, because $\Delta_1 \equiv 1 \pmod{p_1}$ and $\Delta_2 \equiv 0 \pmod{p_1}$. The stated bound suffices to ensure $\|\Delta_1 m + \Delta_2\eta(c)\|_\infty < q/2$ for all m.

Fresh encryption. Let the public key be $(b, -a)$ with $b = as + CRT_{p_1,p_2}(0, e)$ and let a fresh ciphertext use $u, e', e'' \leftarrow \chi$:

$$c_0 = bu + CRT_{p_1,p_2}(m, e'), \qquad c_1 = -au + CRT_{p_1,p_2}(0, e'').$$

A direct cancellation gives

$$\eta_{\text{fresh}} = eu + e' + e''s, \qquad v(c) = \Delta_1 m + \Delta_2 \eta_{\text{fresh}}. \tag{6}$$

Using the multiplicative property of the canonical embedding (coordinate-wise multiplication in $\mathbb{C}^n$),

$$\|xy\|_{\text{can}} \le \|x\|_{\text{can}}\,\|y\|_{\text{can}} \qquad (\forall x, y \in R), \tag{7}$$

and the triangle inequality, we obtain with probability $\ge 1 - 3\varepsilon$,

$$\|\eta_{\text{fresh}}\|_{\text{can}} \;\le\; \|e\|_{\text{can}}\|u\|_{\text{can}} + \|e'\|_{\text{can}} + \|e''\|_{\text{can}}\|s\|_{\text{can}} \;\le\; \tau(\varepsilon)^2\sigma^2 n + 2\,\tau(\varepsilon)\sigma\sqrt{n}. \tag{8}$$

If s, u are bounded small secrets (e.g., ternary), simply substitute high-probability bounds on $\|s\|_{\text{can}}, \|u\|_{\text{can}}$ in place of $\tau(\varepsilon)\sigma\sqrt{n}$.

Addition Given $c^{(1)}, c^{(2)}$ encrypting m_1, m_2 with errors η_1, η_2, the sum $c^{(+)}$ has

$$v(c^{(+)}) = \Delta_1(m_1 + m_2) + \Delta_2(\eta_1 + \eta_2), \quad \text{so} \quad \|\eta(c^{(+)})\|_{\text{can}} \le \|\eta_1\|_{\text{can}} + \|\eta_2\|_{\text{can}}.$$

Multiplication (pre-relinearization) Let $c^{(1)}, c^{(2)}$ encrypt m_1, m_2 with errors η_1, η_2. Before relinearization, the degree-2 ciphertext has critical polynomial

$$g(X) = \left(c_0^{(1)} + c_1^{(1)}s\right)\left(c_0^{(2)} + c_1^{(2)}s\right),$$

and hence

$$g(s) = \big(\Delta_1 m_1 + \Delta_2\eta_1\big)\big(\Delta_1 m_2 + \Delta_2\eta_2\big) = \Delta_1^2 m_1 m_2 \;+\; \underbrace{\Delta_1\Delta_2(m_1\eta_2 + m_2\eta_1) + \Delta_2^2\eta_1\eta_2}_{\text{new error term}}. \tag{9}$$

Consequently, the *pre-relinearization* centered error satisfies

$$\|\eta_{\text{mult}}^{\text{pre}}\|_{\text{can}} \;\le\; |\Delta_1\Delta_2|\big(\|m_1\|_{\text{can}}\|\eta_2\|_{\text{can}} + \|m_2\|_{\text{can}}\|\eta_1\|_{\text{can}}\big) + |\Delta_2|^2\|\eta_1\|_{\text{can}}\|\eta_2\|_{\text{can}}. \tag{10}$$

(Here we use (7).)

Relinearization (key switching) Let $g(s) = d_0 + d_1 s + d_2 s^2$ be the degree-2 ciphertext prior to key switching. Assume a gadget base $B \ge 2$ and length $\ell = \lceil \log_B q \rceil$, and an evaluation key

$$\mathrm{EK}_i = (a_i,\; b_i = a_i s + e_i - B^i s^2) \in R_q^2, \qquad e_i \leftarrow \chi \text{ independent.}$$

Let $g_B^{-1}(d_2) = (\delta_0, \dots, \delta_{\ell-1}) \in R^\ell$ be the coefficient-wise base-B decomposition. The relinearized ciphertext replaces $d_2 s^2$ by $\sum_i \delta_i\,\mathrm{EK}_i$, injecting error $\sum_i \delta_i e_i$. Define the gadget operator constant

$$\Gamma_B := \sup_{x \ne 0} \frac{\|g_B^{-1}(x)\|_{2,\text{can}}}{\|x\|_\infty} \qquad \text{(depends only on } B, \ell \text{ and } R).$$

By Cauchy–Schwarz and independence, with probability $\geq 1-\varepsilon$,

$$\Big\| \sum_{i=0}^{\ell-1} \delta_i e_i \Big\|_{\mathrm{can}} \;\leq\; \|g_B^{-1}(d_2)\|_{2,\mathrm{can}} \cdot \Big(\sum_{i=0}^{\ell-1} \|e_i\|_{\mathrm{can}}^2 \Big)^{1/2} \;\leq\; \Gamma_B \, \|d_2\|_\infty \cdot \tau(\varepsilon)\, \sigma \sqrt{n\ell}. \quad (11)$$

Therefore, if $\eta_{\mathrm{mult}}^{\mathrm{pre}}$ is the centered error before relinearization and $v_{\mathrm{mult}}^{\mathrm{pre}} = \Delta_1^2 m_1 m_2 + \Delta_2 \eta_{\mathrm{mult}}^{\mathrm{pre}}$, the *post-relinearization* centered error satisfies

$$\|\eta_{\mathrm{mult}}^{\mathrm{post}}\|_{\mathrm{can}} \;\leq\; \|\eta_{\mathrm{mult}}^{\mathrm{pre}}\|_{\mathrm{can}} \;+\; |\Delta_2|\, \Gamma_B \, \|d_2\|_\infty \cdot \tau(\varepsilon)\, \sigma \sqrt{n\ell}. \quad (12)$$

Modulus switching / rescaling To keep the message scaling invariant after multiplication, we follow [5, 9] and *modulus switch* from q to $q' = \lfloor q/s \rfloor$ with a scale factor $s \approx \Delta_1$. Let $MS_{q\to q'}$ be coefficient-wise rounding of $(q'/q)\cdot(\cdot)$. For any $x \in R_q$ with centered lift $\tilde{x} \in R$,

$$MS_{q\to q'}(x) \text{ has centered lift } \;\widetilde{x'} = \frac{q'}{q}\, \tilde{x} + r, \quad \text{with} \quad \|r\|_\infty \leq \tfrac{1}{2}.$$

Applying (2), we get

$$\|\widetilde{x'}\|_{\mathrm{can}} \;\leq\; \frac{q'}{q}\, \|\tilde{x}\|_{\mathrm{can}} + \kappa_{\mathrm{can}\leftarrow\infty} \frac{\sqrt{n}}{2}. \quad (13)$$

In particular, if we apply modulus switching to $v_{\mathrm{mult}}^{\mathrm{post}}$ with $s \approx \Delta_1$, then the message coefficient scales from Δ_1^2 back to $\approx \Delta_1$ and the centered error scales by the same factor, plus the additive rounding term in (13).

Noise recurrences and budget. Let $\rho(c) := \|\eta(c)\|_{\mathrm{can}}$ denote the canonical centered CRT-error norm. Let $M := \max\{\|m\|_{\mathrm{can}} \colon m \in R_{p_1}\}$ (worst-case plaintext magnitude in canonical norm). Then, combining the bounds above:

- **Fresh:** with probability $\geq 1 - 3\varepsilon$,
$$\rho_{\mathrm{fresh}} \;\leq\; \tau(\varepsilon)^2 \sigma^2 n + 2\,\tau(\varepsilon)\sigma\sqrt{n}.$$
- **Addition:**
$$\rho(c^{(1)} + c^{(2)}) \;\leq\; \rho(c^{(1)}) + \rho(c^{(2)}).$$
- **Multiplication (before relinearization):**
$$\rho_{\mathrm{mult}}^{\mathrm{pre}} \;\leq\; |\Delta_1 \Delta_2| \cdot \big(M\, \rho(c^{(1)}) + M\, \rho(c^{(2)}) \big) \;+\; |\Delta_2|^2 \cdot \rho(c^{(1)})\, \rho(c^{(2)}).$$
- **Relinearization (additive term):**
$$\rho_{\mathrm{mult}}^{\mathrm{post}} \;\leq\; \rho_{\mathrm{mult}}^{\mathrm{pre}} \;+\; |\Delta_2|\, \Gamma_B \, \|d_2\|_\infty \cdot \tau(\varepsilon)\, \sigma \sqrt{n\ell}.$$
- **Modulus switching by $s \approx \Delta_1$:** writing $q' = \lfloor q/s \rfloor$,
$$\rho' \;\leq\; \frac{q'}{q}\, \rho_{\mathrm{mult}}^{\mathrm{post}} \;+\; \kappa_{\mathrm{can}\leftarrow\infty} \frac{\sqrt{n}}{2}.$$

For decryption, Lemma 1 gives a *uniform* condition that is convenient for parameter setting:

$$\kappa_{\infty\leftarrow\mathrm{can}} \cdot |\Delta_2| \cdot \rho(c) \;<\; \frac{q}{2} - M_\infty. \quad (14)$$

This motivates the following:

Definition 3 (CRT noise budget). *For a ciphertext c at modulus q, define*

$$\mathit{Budget}(c) \ := \ \log_2\left(\frac{q/2 - M_\infty}{\kappa_{\infty\leftarrow\mathrm{can}}\,|\Delta_2|\,\rho(c)}\right).$$

Correctness holds whenever $\mathit{Budget}(c) > 0$. Each homomorphic operation updates $\rho(\cdot)$ according to the recurrences above, and modulus switching contracts both message scale and $\rho(\cdot)$ by a factor $\approx q'/q$ up to an additive rounding term.

Remarks on instantiation. (i) The constants $\kappa_{\infty\leftarrow\mathrm{can}}, \kappa_{\mathrm{can}\leftarrow\infty}$ depend only on R and can be bounded tightly for power-of-two cyclotomics [17]. (ii) If s, u are bounded small secrets, replace the $\tau(\varepsilon)\sigma\sqrt{n}$ factors by the corresponding high-probability bounds for $\|s\|_{\mathrm{can}}, \|u\|_{\mathrm{can}}$; this typically improves the fresh noise bound (8). (iii) Choosing the modulus chain so that each multiplication is followed by modulus switching with factor $s \approx \Delta_1$ (and relinearization base B with moderate Γ_B) yields the same asymptotic noise behavior as in BGV/BFV: additive growth under addition, multiplicative/cross-term growth under multiplication, an additive $\tilde{O}(\sigma\sqrt{n\ell})$ from relinearization, and contraction by q'/q under modulus switching.

4 Security Analysis

This section presents the security foundations of our cryptosystem. We introduce a new hardness assumption, the *Chinese Remainder Theorem Ring Learning With Errors* (CRT-RLWE) problem, which extends the standard RLWE problem by incorporating structured noise via CRT encoding. This adaptation supports improved batching and efficiency while retaining the quantum-resistant properties of RLWE.

We first recall the standard RLWE problem and then define its CRT-based analogue.

Definition 4 (Decision Ring-LWE Problem). *Let $f(X) = X^n + 1$ (with n a power of two) and define the cyclotomic ring $\mathcal{R}_q = \mathbb{Z}_q[X]/\langle f(X)\rangle$. Fix a noise distribution χ over $\mathcal{R}_q$ and draw a secret $s \xleftarrow{\$} \mathcal{R}_q$. The* Ring-LWE distribution *$\mathcal{A}_{s,\chi}$ on $\mathcal{R}_q \times \mathcal{R}_q$ is*

$$\mathcal{A}_{s,\chi} := \Big\{ \big(a,\ a \cdot s + e\big) \ \Big|\ a \xleftarrow{\$} \mathcal{R}_q,\ e \xleftarrow{\$} \chi \Big\}.$$

For any probabilistic polynomial-time (PPT) distinguisher $\mathcal{D}$ we define its advantage *in the average-case decision Ring-LWE problem as*

$$\mathit{Adv}_{\mathcal{D}}^{\mathrm{RLWE}_{n,q,\chi}} := \Big| \Pr_{s,a,e}\big[\mathcal{D}\big(a, a\cdot s + e\big) = 1\big] - \Pr_{a,b\leftarrow\mathcal{R}_q^2}\big[\mathcal{D}(a,b) = 1\big] \Big|.$$

The decision Ring-LWE problem $\mathrm{RLWE}_{n,q,\chi}$ *is said to be* hard *if $\mathit{Adv}_{\mathcal{D}}^{\mathrm{RLWE}_{n,q,\chi}}$ is negligible in the security parameter for every PPT distinguisher $\mathcal{D}$.*

In other words we can write

$$\mathrm{Adv}_{\mathcal{D}}^{\mathrm{RLWE}_{n,q,\chi}} \leq \epsilon_{RLWE}(\lambda)$$

Definition 5 (CRT-RLWE Distribution). *Let s be a secret in the ring $\mathcal{R}_q$, and let χ be a noise distribution over $K_{\mathbb{R}}$. A sample from the Chinese Remainder Theorem-based Ring-LWE (CRT-RLWE) Distribution, denoted as $C_{s,\chi}$, is a pair:*

$$(\mathbf{a}, \mathbf{b}), \quad \textit{where} \quad \mathbf{b} = \mathbf{a} \cdot \mathbf{s} + CRT_{p_1,p_2}(\mathbf{0}, \mathbf{e})$$

Here, $\mathbf{a} \leftarrow \mathcal{R}_q$ *is chosen uniformly at random, and* $\mathbf{e} \leftarrow \chi$*. The CRT-RLWE distribution is formally defined as:*

$$C_{s,\chi} = \{(\mathbf{a}, \mathbf{b} = \mathbf{a} \cdot \mathbf{s} + CRT_{p_1,p_2}(\mathbf{0}, \mathbf{e})) \mid \mathbf{a} \leftarrow \mathcal{R}_q, \quad \mathbf{e} \leftarrow \chi\}$$

Definition 6 (Decision CRT-RLWE Problem). *The* Decision CRT-RLWE problem*, denoted* $\text{CRT-DLWE}_{q,\chi}$*, is defined as follows:*

For any probabilistic polynomial-time (PPT) distinguisher $\mathcal{D}$*, the advantage in solving the decision CRT-RLWE problem is:*

$$Adv_{\mathcal{D}}^{\text{CRT-DLWE}_{q,\chi}} := \left| \Pr_{s,\mathbf{a},\mathbf{e}}[\mathcal{D}(\mathbf{a}, \mathbf{a} \cdot s + \text{CRT}_{p_1,p_2}(0, \mathbf{e})) = 1] - \Pr_{\mathbf{a},\mathbf{b} \xleftarrow{\$} \mathcal{R}_q^2}[\mathcal{D}(\mathbf{a}, \mathbf{b}) = 1] \right|$$

We say that the decision CRT-RLWE problem is hard *if* $Adv_{\mathcal{D}}^{\text{CRT-DLWE}_{q,\chi}}$ *is negligible for all PPT distinguishers* $\mathcal{D}$*.*

In other words we can write

$$\text{Adv}_{\mathcal{D}}^{\text{CRT-DLWE}_{q,\chi}} \leq \epsilon_{CRT-RLWE}(\lambda)$$

In Lemma 2 and 3 we provide a formal reduction showing that solving the Decision CRT-RLWE problem efficiently implies the ability to solve the Decision RLWE problem efficiently. This establishes that the hardness of the Decision RLWE problem at most the hardness of the Decision CRT-RLWE problem.

Lemma 2. *Let* $s \in \mathcal{R}_q$ *be a secret and* χ *a noise distribution over* $\mathcal{R}_q$*. Then any sample* $(\mathbf{a}, \mathbf{b})$ *from the standard RLWE distribution* $A_{s,\chi}$

$$\mathbf{b} = \mathbf{a} \cdot \mathbf{s} + \mathbf{e} \in \mathcal{R}_q, \quad \mathbf{e} \leftarrow \chi$$

can be efficiently transformed into a sample $(\mathbf{a}_{CRT}, \mathbf{b}_{CRT})$ *from the CRT-RLWE distribution* $C_{s,\chi}$

$$\mathbf{b}_{CRT} = \mathbf{a}_{CRT} \cdot \mathbf{s} + CRT_{p_1,p_2}(0, \mathbf{e})$$

using a polynomial-time transformation.

Proof. Given an RLWE sample $(\mathbf{a}, \mathbf{b})$, where $\mathbf{a} \in \mathcal{R}_q, \mathbf{b} = \mathbf{a} \cdot \mathbf{s} + \mathbf{e} \mod q$, we define a CRT embedding $\mathbf{b}_{\text{CRT}} \in \mathcal{R}_q$ as follows:

$$\mathbf{b}_{\text{CRT}} = \Delta_2 \cdot \mathbf{b} \mod q$$

Substituting $\mathbf{b} = \mathbf{a} \cdot \mathbf{s} + \mathbf{e}$, we obtain:

$$\mathbf{b}_{\text{CRT}} = (\Delta_2 \cdot \mathbf{a}) \cdot \mathbf{s} + \Delta_2 \cdot \mathbf{e} = (\Delta_2 \cdot \mathbf{a}) \cdot \mathbf{s} + CRT_{p_1,p_2}(\mathbf{0}, \mathbf{e}).$$

If $\mathbf{e}$ is small enough such that $\|\Delta_2 \cdot \mathbf{e}\| < q/2$, then we can express:

$$\Delta_2 \cdot \mathbf{e} \mod q = CRT_{p_1,p_2}(\mathbf{0}, \mathbf{e}) \mod q.$$

So we can write

$$\mathbf{b}_{\text{CRT}} = (\Delta_2 \cdot \mathbf{a}) \cdot \mathbf{s} + CRT_{p_1,p_2}(\mathbf{0}, \mathbf{e}).$$

Further, we define

$$\mathbf{a}_{\mathrm{CRT}} = \Delta_2 \cdot \mathbf{a}$$

Thus, this transformation correctly simulates the CRT-RLWE problem instance:

$$(\mathbf{a}_{\mathrm{CRT}}, \mathbf{b}_{\mathrm{CRT}}) = (\Delta_2 \cdot \mathbf{a}, \mathbf{b}_{\mathrm{CRT}}).$$

The transformation $\mathbf{b} \to \mathbf{b}_{\mathrm{CRT}}$ preserves the RLWE problem's structure within the CRT-RLWE framework:

$$\mathbf{b}_{\mathrm{CRT}} = \mathbf{a}_{\mathrm{CRT}} \cdot \mathbf{s} + CRT_{p_1,p_2}(\mathbf{0}, \mathbf{e}) \mod q \iff \mathbf{b} - \mathbf{a} \cdot \mathbf{s} = \mathbf{e} \mod q.$$

Specifically:

- The CRT embedding maps $\mathbf{b}$ to $\mathbf{b}_{\mathrm{CRT}}$, embedding the RLWE noise $\mathbf{e}$ into the CRT representation $CRT_{p_1,p_2}(\mathbf{0}, \mathbf{e})$, which matches the CRT-RLWE formulation.
- The secret $\mathbf{s}$ remains identical between the RLWE and CRT-RLWE formulations.

This demonstrates that samples from $A_{s,\chi}$ can be efficiently converted to samples from $C_{s,\chi}$, via a polynomial-time transformation.

Lemma 3 (Reduction from Decision RLWE to Decision CRT-RLWE). *Let q be a modulus, $\mathcal{R}_q$ a quotient ring, and χ a noise distribution. If there exists an efficient algorithm that solves the Decision CRT-RLWE problem CRT-DRLWE$_{q,\chi}$ with non-negligible advantage, then there exists an efficient algorithm that solves the Decision RLWE problem R-DLWE$_{q,\chi}$ with non-negligible advantage.*

Proof. We construct a reduction from the Decision RLWE problem to the Decision CRT-RLWE problem by transforming an RLWE instance into a CRT-RLWE instance while preserving the underlying hardness assumption.

Step 1: Input Transformation Given an instance of the Decision RLWE problem, we are provided with a tuple $(\mathbf{a}, \mathbf{b})$ sampled from $A_{s,\chi}$, where $\mathbf{s}$ is sampled from $\mathcal{R}_q$ uniformly and:

$$\mathbf{b} = \mathbf{a} \cdot \mathbf{s} + \mathbf{e} \mod q.$$

Our goal is to transform this instance into an instance of the Decision CRT-RLWE problem based on the CRT-RLWE distribution $C_{s,\chi}$. By applying the transformation described in Lemma 2, we construct the corresponding CRT-RLWE sample as follows:

$$(\mathbf{a}_{\mathrm{CRT}}, \mathbf{b}_{\mathrm{CRT}}) = (\Delta_2 \cdot \mathbf{a}, \Delta_2 \cdot \mathbf{b}) = (\Delta_2 \cdot \mathbf{a}, \quad \Delta_2 \cdot \mathbf{a} \cdot \mathbf{s} + CRT_{p_1,p_2}(\mathbf{0}, \mathbf{e}))$$

By lemma 2 we can conclude that $(\mathbf{a}_{\mathrm{CRT}}, \mathbf{b}_{\mathrm{CRT}})$ is a valid CRT-RLWE sample from $C_{s,\chi}$.

By assumption, we have access to an efficient oracle $\mathcal{B}$ that can distinguish between valid CRT-RLWE samples and uniform random samples with non-negligible advantage.

Step 2: Oracle Query. We give $(\mathbf{a}_{\mathrm{CRT}}, \mathbf{b}_{\mathrm{CRT}})$ to the CRT-RLWE oracle $\mathcal{B}$, which returns:

- True if the pair is a valid CRT-RLWE sample;
- False if the pair is uniformly random.

Decision Rule. We interpret $\mathcal{B}$'s output as the decision for the original RLWE problem:

- If $\mathcal{B}$ returns True, we output "RLWE".
- If $\mathcal{B}$ returns False, we output "Uniform".

Correctness. The transformation from RLWE to CRT-RLWE preserves distributional structure. In the real case, the CRT embedding of the noise ensures the CRT-RLWE form is satisfied; in the random case, uniform randomness is preserved since CRT is a ring isomorphism. Therefore, any non-negligible advantage of $\mathcal{B}$ in distinguishing CRT-RLWE samples translates directly to an advantage in distinguishing RLWE samples.

Conclusion. Hence, the existence of an efficient solver for Decision CRT-RLWE implies the existence of an efficient solver for Decision RLWE.

This reduction demonstrates that an efficient solver for the Decision CRT-RLWE problem can be used to solve the Decision RLWE problem. Consequently, the hardness of the Decision RLWE problem is at most that of the Decision CRT-RLWE problem. This equivalence supports the security of the proposed cryptographic scheme by leveraging the well-established hardness of the RLWE problem.

Lemma 4 (IND-CPA Security). *The proposed CRT-FHE cryptosystem is IND-CPA secure under the hardness of the Decision CRT-RLWE problem.*

Proof. We prove the theorem using a sequence of games $G_0 \to G_1 \to G_2$, and bound the adversary's advantage via the triangle inequality.

Game G_0: Real IND-CPA Game. In this game, the adversary $\mathcal{A}$ receives:

- a public key $\mathsf{pk} = (\mathbf{pk}_0, \mathbf{pk}_1) = (\mathbf{a}\cdot\mathbf{s} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}), -\mathbf{a}) \in \mathcal{R}_q^2$, where $\mathbf{a}, \mathbf{s} \xleftarrow{\$} \mathcal{R}_q$, $\mathbf{e} \xleftarrow{\$} \chi$
- a challenge ciphertext encrypting a plaintext $\mathbf{m}_b \in R_{p_1}$, where $b \in \{0, 1\}$ is chosen uniformly at random.

Encryption proceeds by sampling $\mathbf{u}, \mathbf{e}', \mathbf{e}'' \xleftarrow{\$} \chi$ and computing:

$$\mathbf{c}_0 = \left[\mathbf{pk}_0 \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m}_b, \mathbf{e}')\right]_q$$
$$\mathbf{c}_1 = \left[\mathbf{pk}_1 \cdot \mathbf{u} + \mathrm{CRT}_{p_1,p_2}(0, \mathbf{e}'')\right]_q$$

The adversary outputs a guess b'.

Define:

$$\mathrm{Adv}_{\mathcal{A}}^{G_0} := \left|\Pr[\mathcal{A} \text{ outputs } b' = b] - \frac{1}{2}\right|$$

Game G_1: Replace Public Key with Uniform. In this game, we sample $(\mathbf{pk}_0, \mathbf{pk}_1) \xleftarrow{\$} \mathcal{R}_q^2$ uniformly at random. The challenge ciphertext is still computed using the same encryption procedure.

We want to prove that Game G_1 is indistinguishable from Game G_0. Suppose there exists any PPT adversary $\mathcal{A}$ which aims to distinguish Game G_0 and Game G_1 with non-negligible advantage. Then we can construct a distinguisher $\mathcal{D}$ aims to break CRT-DRLWE$_{n,q,\chi}$. $\mathcal{D}$ receives a sample $(pk_1, pk_2) \in \mathcal{R}_q^2$, then runs $\mathcal{A}$ with (pk_1, pk_2)

as the public key. Here $\mathcal{D}$ can perfectly simulate Game G_0 or Game G_1 depending on whether its input samples are CRT-RLWE or uniform (respectively). Therefore,$\mathcal{D}$ and $\mathcal{A}$ have equal distinguishing advantages. Because $\mathcal{D}$'s advantage must be negligible by CRT-RLWE assumption (Definition 6), so is $\mathcal{A}$'s.

In other words, we can express the advantage of $\mathcal{A}$ winning the Game G_0 and Game G_1 using the advantage of $\mathcal{D}$ which is negligible in the security parameter λ.

$$\mathrm{Adv}_{\mathcal{D}}^{\text{CRT-RLWE}_{n,q,\chi}} = |\Pr[\mathcal{A} \text{ wins in } G_0] - \Pr[\mathcal{A} \text{ wins in } G_1]|$$

By assumption (hardness of Decision CRT-RLWE), for all PPT $\mathcal{D}$,

$$\mathrm{Adv}_{\mathcal{D}_1}^{\text{CRT-RLWE}} \leq \epsilon_1(\lambda)$$

Thus:

$$|\Pr[\mathcal{A} \text{ wins in } G_0] - \Pr[\mathcal{A} \text{ wins in } G_1]| \leq \epsilon_{CRT-RLWE}(\lambda)$$

Game G_2: Replace Ciphertext with Uniform. In this game, we again use a public key sampled uniformly from $\mathcal{R}_q^2$, as in G_1. We now also replace the ciphertext with uniformly random elements $(\mathbf{c}_0, \mathbf{c}_1) \xleftarrow{\$} \mathcal{R}_q^2$, independently of the public key.

We want to prove that Game G_1 and Game G_2 are indistinguishable from $\mathcal{A}$. The ciphertext generated by a public key which is chosen uniformly random from $\mathcal{R}_q^2$ has a distribution like the following:

$$\mathcal{E}_{s,\chi}\{(\mathbf{a}, \mathbf{a}\cdot\mathbf{s} + \mathrm{CRT}_{p_1,p_2}(\mathbf{m},\mathbf{e})) = ((\mathbf{a}, \mathbf{a}\cdot\mathbf{s} + \mathrm{CRT}_{p_1,p_2}(\mathbf{0},\mathbf{e}) + \Delta m)\},$$

where $\mid \mathbf{a} \xleftarrow{\$} \mathcal{R}_q, \mathbf{e} \xleftarrow{\$} \chi, \mathbf{m} \xleftarrow{\$} R_{p_1}, \Delta = p_2 \cdot [p_2^{-1}]_{p_1}$

Then the $\mathcal{E}_{s,\chi}$ is indistinguishable from uniform distribution since all the elements are CRT-RLWE samples adding any fixed message $\Delta m \in \mathcal{R}_q$, which preserves the uniformity. Moreover, Game G_1 and Game G_2 are indistinguishable for any fixed m and Game G_2 does not depend on m at all, therefore, the two games for $m = 0, 1$ are indistinguishable.

In other words, we can express the advantage of $\mathcal{A}$ winning the Game G_1 and Game G_2 using the advantage of $\mathcal{D}_\in$ a PPT adversary that distinguishes encryptions under $\mathcal{E}_{s,\chi}$ (defined below) from uniform.

The advantage of $\mathcal{D}_2$ is the same as $\mathcal{D}_1$ which is CRT-RLWE distinguisher.

$$|\Pr[\mathcal{A} \text{ wins in } G_1] - \Pr[\mathcal{A} \text{ wins in } G_2]| = \mathrm{Adv}_{\mathcal{D}_2}^{\mathcal{E}_{s,\chi}} = \mathrm{Adv}_{\mathcal{D}_1}^{\text{CRT-RLWE}} \leq \epsilon_{CRT-RLWE}(\lambda)$$

Game G_2 Analysis. In Game G_2, the ciphertext is sampled uniformly and independently of the plaintext bit b. Therefore, regardless of $\mathcal{A}$'s strategy,

$$\Pr[\mathcal{A} \text{ wins in } G_2] = \frac{1}{2}$$

Final Bound. Combining all bounds via triangle inequality:

$$\begin{aligned}\left|\Pr[\mathcal{A} \text{ wins in } G_0] - \frac{1}{2}\right| &= |\Pr[\mathcal{A} \text{ wins in } G_0] - \Pr[\mathcal{A} \text{ wins in } G_2]| \\ &\leq |\Pr[\mathcal{A} \text{ wins in } G_0] - \Pr[\mathcal{A} \text{ wins in } G_1]| \\ &\quad + |\Pr[\mathcal{A} \text{ wins in } G_1] - \Pr[\mathcal{A} \text{ wins in } G_2]| \\ &\leq \epsilon_{CRT-RLWE}(\lambda) + \epsilon_{CRT-RLWE}(\lambda)\end{aligned}$$

Hence, the advantage of any PPT adversary $\mathcal{A}$ in the IND-CPA game is negligible under the Decision CRT-RLWE assumption, and the CRT-FHE cryptosystem is IND-CPA secure.

5 Performance Analysis

5.1 Experimental Setup

All experiments were run on a MacBook Pro (2022) with an Apple M2 SoC (4 performance + 4 efficiency cores), 24 GB RAM, macOS Sequoia 15.2. The CRT-FHE codebase was compiled with `rustc 1.85.1`; OpenFHE (BFV/CKKS) was compiled with `clang 17.0.0`. We used Criterion.rs for CRT-FHE and Google Benchmark for OpenFHE, both configured for single-threaded runs. Each benchmark performed one warm-up followed by 50 sampling iterations; we report point estimates together with 95% confidence intervals as produced by the respective harnesses. Unless otherwise noted, all measurements are wall-clock latencies; throughput is computed as *slots per millisecond* (slots/ms).

Fair-comparison notes. Different harnesses (Criterion vs. Google Benchmark) can yield slightly different variance models; we therefore focus on (i) medians and (ii) relative effects that are large compared to reported CIs. We also report per-slot throughput to normalize across schemes with different effective slot counts or polynomial degrees.

5.2 Cryptographic Parameters

For CRT-FHE and BFV we used polynomial degree n=8192 (yielding 8192 SIMD slots) and matched ciphertext modulus ladders $Q = \{35{,}175{,}245{,}135{,}873,\ 35{,}156{,}991{,}246{,}337\}$. The plaintext modulus was t=65537; the CRT embedding used an auxiliary parameter p_1=3. For CKKS in OpenFHE we followed a common configuration with n=16384 (effective 8192 complex slots) and $Q = \{1{,}152{,}921{,}504{,}606{,}748{,}673,\ 281{,}474{,}977{,}595{,}393\}$. We report per-slot throughput to partially normalize across these degree choices; a full security/precision alignment (e.g., equal BKZ cost or CKKS scale/precision targets) is left to future work.

5.3 Operation Benchmarks

Table 1. Operation benchmarks with latency (ms), throughput (slots/ms), and relative speed of CRT-FHE vs. BFV/CKKS. For "CRT vs X", values > 1 mean CRT-FHE is faster; < 1 means slower. Best (lowest) latency per row is bold.

Operation	**Latency (ms) ↓**			**CRT vs X (×)**		**Throughput (slots/ms) ↑**		
	CRT-FHE	**BFV**	**CKKS**	BFV	CKKS	**CRT-FHE**	**BFV**	**CKKS**
Encrypt	**1.912**	1.984	2.979	1.04	1.56	**4,283**	4,128	2,750
Add	**0.0130**	0.0249	0.0545	1.92	4.19	**630,154**	328,915	150,312
Add (in-place)	**0.0070**	0.0186	0.0400	2.66	5.71	**1,170,286**	440,430	204,800
Subtract	**0.189**	—	—	—	—	**43,333**	—	—
Subtract (in-place)	**0.184**	—	—	—	—	**44,512**	—	—
Multiply + Relin	2.616	**2.193**	2.211	0.84	0.85	3,130	**3,733**	3,705

Relative effects (point estimates).

- **Encryption.** CRT-FHE is $\sim$3.6% faster than BFV (1.912 vs. 1.984 ms; $1.98/1.91 \approx 1.04\times$) and $\sim 1.56\times$ faster than CKKS (2.979 ms).
- **Addition.** CRT-FHE outperforms BFV by $\sim 1.9\times$ (0.0249/0.0130) and CKKS by $\sim 4.2\times$ (0.0545/0.0130).
- **In-place addition.** CRT-FHE outperforms BFV by $\sim 2.66\times$ (0.0186/0.0070) and CKKS by $\sim 5.7\times$ (0.0400/0.0070).
- **Multiplication + relinearization.** CRT-FHE trails BFV by $\sim$19% (2.616 vs. 2.193 ms) and CKKS by $\sim$18% (2.616 vs. 2.211 ms).

Throughput (slots/ms) shows the same qualitative picture and partially compensates for CKKS's larger polynomial degree.

5.4 Interpretation and Discussion

Additive path. CRT-FHE achieves large gains on additions (both standard and in-place). These gains are consistent with (i) a leaner ciphertext layout and (ii) optimized modular add/sub primitives that avoid wide divisions in the inner loop (see §5.6). Per-slot throughput indicates the advantage persists even when normalizing for slot count.

Multiplicative path. On Multiply+Relin, OpenFHE's BFV/CKKS are faster on this hardware/parameter point. Two likely drivers are (a) highly tuned key-switch/relinearization kernels (including gadget decomposition and RNS base conversions) and (b) CKKS/BFV's mature NTT pipelines. Our CRT-FHE implementation has room to close this gap with deeper tuning of key-switch primitives and memory layout.

Encryption. The small advantage over BFV and the larger advantage over CKKS align with the degree difference (CKKS at n=16384) and with CRT-FHE's lightweight public-key structure. The per-slot normalization corroborates this trend.

5.5 Threats to Validity

- **Security/precision parity.** We did not equalize concrete security (e.g., BKZ cost) across schemes, nor CKKS precision targets; results are microbenchmarks, not a security-normalized bake-off.
- **Degree mismatch.** CKKS used n=16384 vs. n=8192 for CRT-FHE/BFV, potentially biasing raw latencies; we therefore also report per-slot throughput.
- **Harness/ABI differences.** CRT-FHE used Rust + Criterion; BFV/CKKS used C++ + Google Benchmark. Small systemic differences (allocator, inlining, exception model) may affect tails.
- **Single-threading on big.LITTLE.** Runs are single-threaded but not affinitized in our harness; occasional migration between performance/efficiency cores could introduce variance on macOS.

5.6 Implementation Notes

Key optimizations in CRT-FHE include:

- **Division-free modular arithmetic.** Barrett-style reduction tailored to Apple M2 avoids 128-bit divisions on hot paths.

- **NTT pipeline.** Precomputed twiddle tables, lazy modular reduction, and cache-aware butterfly scheduling reduce L1/L2 pressure.
- **Core-local ILP.** Loop unrolling tuned for M2's out-of-order cores improves instruction-level parallelism even without explicit SIMD.

All schemes use RNS for large moduli. Our CRT-FHE uses two $\sim$45-bit residues plus a $\sim$60-bit special prime for preconditioning, which minimizes carry propagation in mixed-radix steps.

Takeaway. At the tested parameter points, CRT-FHE delivers substantial advantages on the additive path and competitive encryption latency, while BFV/CKKS retain an edge on multiply+relinearize. We expect further gains from focused tuning of relinearization and base-conversion routines in CRT-FHE, and from a security-normalized comparison across matched modulus chains and degrees.

6 Conclusion

In this paper, we have introduced a novel Fully Homomorphic Encryption (FHE) scheme that leverages the Chinese Remainder Theorem (CRT) to address the critical issue of noise accumulation in homomorphic computations. By uniquely encoding the plaintext with noise, the proposed scheme effectively prevents the noise from growing to the point where it corrupts the plaintext, thereby facilitating smooth computations on encrypted data without the need for bootstrapping. This advancement is significant, as it allows for a predefined number of modular operations on encrypted data, enhancing both efficiency and practicality.

The evolution of FHE has been marked by notable milestones since its conceptual inception by Rivest in the 1970s and its formal realization by Gentry in the late 2000s. While numerous efforts have been made to develop FHE schemes using CRT, these have often fallen short due to vulnerabilities such as chosen plaintext attacks (CPA). Our proposed scheme addresses these vulnerabilities by incorporating random errors into the encryption process in a controlled manner. These errors are designed such that, during homomorphic operations, their growth does not interfere with the integrity of the plaintext, unlike in traditional LWE-based schemes.

The key contributions of the proposed scheme can be summarized as follows:

- Noise Management: By encoding the plaintext with noise in a novel way, we ensure that the noise does not overflow and corrupt the plaintext, enabling reliable homomorphic computations.
- CRT Utilization: We harness the power of the Chinese Remainder Theorem to construct a robust FHE scheme that supports a set number of modular operations without necessitating bootstrapping.
- Security Enhancement: the proposed scheme is resilient to chosen plaintext attacks, thereby improving the security profile compared to previous attempts utilizing CRT.

This research represents a significant step forward in the development of efficient and secure FHE schemes. By overcoming the traditional limitations associated with noise management and CPA vulnerabilities, the proposed scheme opens up new possibilities for practical applications of FHE in fields such as secure data processing, privacy-preserving computations, and encrypted machine learning. Future work will focus on further optimizing the performance and expanding the range of supported operations to enhance the scheme's applicability in diverse real-world scenarios.

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